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Enhancement of the production of bio-aromatics from bamboo pyrolysis: Wet torrefaction pretreatment coupled with catalytic fast pyrolysis

Zhouyang Hu [a,1], Liang Zhu [a,1], Hongyi Cai [b], Ming Huang [a], Jie Li [a], Bo Cai [a], Dengyu Chen [c], Lingjun Zhu [d], Youyou Yang [e,*], Zhongqing Ma [a,*]

[a] College of Chemistry and Materials Engineering, Zhejiang A & F University, Hangzhou, Zhejiang 311300, China
[b] People's Government of Yaoshan village, Chun'an County, Hangzhou, Zhejiang 311703, China
[c] College of Materials Science and Engineering, Nanjing Forestry University, Nanjing City, Jiangsu Province 210037, China
[d] State Key Laboratory of Clean Energy Utilization, Zhejiang University, Hangzhou, Zhejiang 310027, China
[e] College of Humanities and Law, Zhejiang A & F University, Hangzhou, Zhejiang 311300, China

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ABSTRACT

Light aromatics are important organic building blocks in the chemical industry which can be produced by catalytic fast pyrolysis (CFP) of biomass. In this work, wet torrefaction pretreatment (WTP) was employed to improve the quality of bamboo by synergistic effect of deoxygenation and demineralization. Then, CFP was employed to produce bio-aromatics by using zeolite (e.g. HZSM-5, HY, Al-MCM-41, and USY) as catalyst. Results showed that WTP temperature (180–260 °C) had more significant influence on the mass yields of torrefied products compared to WTP duration (30–150 min). The maximum deoxygenation rate was 49.36% at WTP conditions of 260 °C and 150 min, and the maximum demineralization rate followed the order of 96.29% (K) > 94.54% (Na) > 90.33% (Mg) > 89.22% (Ca). Among the five types of zeolite catalyst tested, HZSM-5 (25) was the best catalyst to obtain bio-aromatics due to its unique pore size and reasonable acidity. The maximum yield of aromatics (25.46×10^7 a.u./mg) was obtained at the WTP temperature of 220 °C, biomass-to-catalyst ratio of 3:1, and CFP temperature of 850 °C. Toluene was the more favored monocyclic aromatic hydrocarbon formed during CFP compared to xylene and benzene.

1. Introduction

Benzene, toluene, and xylene (BTX), as the major chemicals in the light aromatics group, are well known as the first-grade organic building blocks in the chemical industry. They are used to manufacture a wide variety of products, such as synthetic fiber, resin, plastic, and solvents [1]. Nowadays, light aromatics are conventionally produced by the catalytic reforming of non-renewable petroleum fractions [2]. However, with the rapid depletion of fossil fuels, development of new technology using renewable feedstock is essential to build a sustainable supply of light aromatics [3]. Lignocellulosic biomass has been considered as the most promising alternative feedstock to fossil fuels due to its renewability, low environmental impact, and abundance [4]. In addition, catalytic fast pyrolysis (CFP) of biomass is a promising technology to produce bio-aromatics [5].

However, the yield of bio-aromatics produced by CFP is usually limited by the drawbacks of biomass, namely, the high oxygen content and high ash content. The high oxygen content (35–45%) in biomass results in a low value of the effective ratio of hydrogen-to-carbon (H/C_{eff}) [6]. The bio-oil derived from fast pyrolysis of biomass is composed of high concentration of oxygenates, but less concentration of valuable hydrocarbon aromatics [7,8]. Moreover, oxygen enrichment in biomass produces low yields of hydrocarbon aromatics even in the CFP process along with the formation of a large amount of coke on the catalyst [9]. The alkali and alkaline earth metals (AAEMs) in ash, such as calcium (Ca), sodium (Na), magnesium (Mg), and potassium (K), act as active catalysts during biomass pyrolysis process. These AAEMs promote the second cracking of the organic compounds in pyrolysis vapor, transforming them into the non-condensable gaseous products, which greatly reduces the yield of bio-oil [7,10–12]. Hwang et al. [13] found that the

* Correspondence to: College of Chemistry and Materials Engineering, Zhejiang A & F University, 666 Wusu Road, Hangzhou City, Zhejiang Province 311300, China.
E-mail addresses: yangyy@zafu.edu.cn (Y. Yang), mazq@zafu.edu.cn (Z. Ma).
[1] Zhouyang Hu# and Liang Zhu#, these two authors contributed equally to this work.

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yield of bio-oil decreased by 17.4% after impregnation of K on poplar wood. Wang et al. [14] reported that the yield of bio-oil decreased by 16% after impregnation of Na on lignin. Wang et al. [15] investigated the effect of inorganic salts on the pyrolysis of cellulose, and reported that the yield of bio-oil decreased by 19.8% and 21.5% after impregnation of Ca and Mg, respectively. Therefore, employing pretreatment technology for deoxygenation and demineralization of biomass prior to CFP is essential to enhance the yield of bio-aromatics.

Torrefaction is a promising pretreatment technology to overcome the aforementioned drawbacks of biomass [16]. According to the different reaction medium, torrefaction pretreatment can be divided into two techniques, namely, dry torrefaction pretreatment (DTP) using the carrier gas (nitrogen or air) as reaction medium and wet torrefaction pretreatment (WTP) using the water solution as reaction medium [17,18]. WTP, also known as hydrothermal pretreatment, can upgrade the properties of biomass in hot-compressed water at 180–260 °C [19,20]. The low-grade biomass feedstock can be upgraded by either one of these two pretreatments (DTP and WTP) in terms of higher energy density and higher H/C_{eff} by removing part of oxygen element. However, compared to DTP, WTP not only removes oxygen element from biomass, but also removes the inorganic elements (e.g., AAEMs) by dissolving them in the water reaction medium [21–23].

In terms of the deoxygenation rate of biomass after WTP, Kambo et al. [24] and Zhang et al. [25] investigated the influence of WTP temperature on the properties of miscanthus and bamboo, and found that the maximum values of the deoxygenation rate were 30.24% and 35.87%, respectively. Stemann et al. [26], Hoekman et al.[27,28], and Nakason et al. [29] found that the oxygen element in wet torrefied biomass was gradually transferred into torrefied gaseous and liquid products at higher WTP temperature. Non-condensable CO_2 and CO were the dominant oxygen carriers in torrefied gaseous product. Organic acids (e.g., formic acid, acetic acid, and lactic acid) and sugars (e.g. xylose, glucose, fructose, and levoglucosan) were the oxygen carriers in torrefied liquid product. The de-ash rate of biomass was normally higher than 50% after WTP, but varied in a small range depending on different species of lignocellulosic biomass, such as 59% for corn stalk [18], 57% for spruce wood [21], and 53% for moso bamboo [30]. Zhang et al. found that the contents of AAEMs decreased with the raising of WTP temperature, the demineralization rates of K, Na, Mg, and Ca were 92.67%, 92.19%, 84.71%, and 78.92% [31]. Results showed that the alkali metals (K and Na) were more easily to be removed by WTP than the alkali earth metals (Mg and Ca) due to their different solubilities in water solution. Therefore, more favorable quality of biomass with less undesirable oxygen element and inorganic components can be obtained after WTP.

Biomass CFP is a promising technology to produce bio-aromatics by employing zeolite as catalyst [5]. During CFP process, the oxygen-containing compounds in pyrolysis vapor convert into hydrocarbon aromatics through the deoxygenation and aromatization reactions [3,32]. Recently, various types of zeolite catalysts (e.g. HZSM-5, H-Y, H-β, Al-MCM-41, SBA-15, USY) have been studied extensively for CFP to improve the yield of bio-aromatics [4,33–35]. Shen et al. [34] investigated the effect of different types of zeolite catalysts (HZSM-5 (25), HZSM-5 (50), HZSM-5 (210), H-β, and USY) on the yield of aromatics during lignin CFP, and found that HZSM-5 (25) presented the highest yield of aromatics due to its relatively large surface area and suitable acidity. Jin et al. [33] found that the yield of aromatics by employing HZSM-5 (25) as catalyst was higher compared to H-Y (25) and H-β (30) catalysts during biomass CFP. This was because HZSM-5 (25) had stronger acidity in the Brönsted acid sites, resulting in higher shape selectivity for the production of aromatics. However, the use of WTP prior to CFP with zeolite as catalyst to enhance the yield of light aromatics from biomass has not been well investigated.

In this work, a novel approach involving WTP coupled with CFP was employed for the enhanced production of bio-aromatics from bamboo pyrolysis (as shown in Fig. 1). First, the influence of WTP temperature and duration on the deoxygenation and deminerilization rate of bamboo was studied. Then, the CFP of bamboo was performed in the analytical Py-GC/MS by employing different types of zeolite catalysts. Finally, the operation parameters of WTP and CFP were optimized.

2. Materials and methods

2.1. Materials

Moso bamboo was harvested from the mountain area of Zhejiang province of China. The outer and inner skins of bamboo were removed by a sander, while the middle part was kept for the torrefaction and pyrolysis experiment. The bamboo was ground into powder, and the particle size between 40 and 60 meshes was kept. Finally, the powder was dried at 105 ºC for 24 h before the wet torrefaction experiment.

2.2. Wet torrefaction pretreatment experiment

The wet torrefaction pretreatment (WTP) of bamboo was performed in a 300 mL autoclave reactor (Xi'an Mogina Instrument Manufacturing Co. Ltd.). The maximum reaction pressure of this reactor was 25 MPa. In each experiment, 10 g bamboo powder and 100 mL deionized water (solid-to-liquid ratio of 1:10) were loaded in the reactor. Nitrogen was introduced into the reactor for at least 30 min until the air was completely removed from the reactor. In order to optimize the WTP parameters, the reactor was heated to varied temperatures of 180, 200, 220, 240, and 260 ºC at the heating rate of 10 ºC/min with different holding times of 30, 60, 90, 120, and 150 min. The reactor was cooled to room temperature in the atmosphere. After cooling of the reactor, the torrefied solid product was collected by the following steps of centrifugation and vacuum filtration. Finally, the torrefied solid product was washed three times using deionized water, and dried at 105 ºC for 24 h

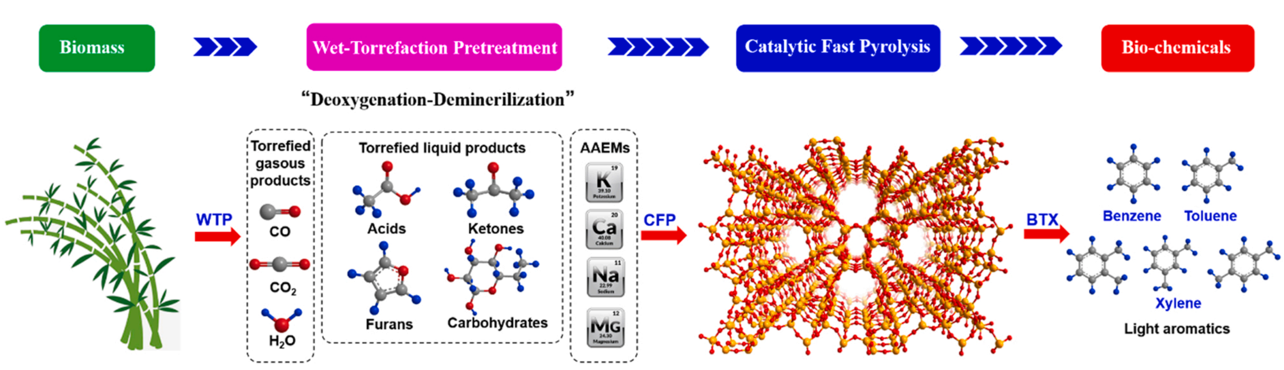

Fig. 1. Combined approach of WTP and CFP to produce bio-aromatics from bamboo.

for further analysis of its basic properties and CFP experiment. The wet torrefied bamboo samples were labelled as BWT-XXX-YYY, where XXX represented the torrefaction temperature, and YYY represented the holding time. For example, B-180–150 represented the torrefied solid product obtained at the temperature of 180 and the holding time of 150 min. The mass yields of torrefied gaseous, liquid, and solid products were calculated according to the Eqs. (1) to (3).

$$Mass\ yield\ of\ the\ solid\ product\ (\%) = \frac{Mass\ of\ solid\ product\ (g)}{Mass\ of\ dried\ feedstock\ (g)} \tag{1}$$

$$Mass\,yield\,of\,the\,liquid\,product\,(\%) = Mass\,yield\,of\,total\,liquid\,and\,solid\,product - mass\,yield\,of\,the\,solid\,product \tag{2}$$

$$Mass\ yield\ of\ the\ gaseous\ product\ (\%) = 100\ \% - mass\ yield\ of\ the\ solid\ and\ liquid\ products\ (\%) \tag{3}$$

2.3. Basic properties of the raw and wet torrefied bamboo samples

The elemental distribution and proximate analysis of the raw and wet torrefied bamboo samples were determined by an automatic proximate analyzer and elemental analyzer. The changes in functional groups and crystallinity of bamboo at varying WTP temperatures were tested by FTIR and XRD. The pyrolysis behaviors of the raw and wet torrefied bamboo samples were performed by the thermogravimetric analyzer. The changes in the inorganic elements, especially the alkali and alkaline earth metals (K, Na, Ca, and Mg) at varying WTP temperature, were tested by an inductively coupled plasma-optical emission spectrometer (ICP-OES 730, Agilent, USA). Exactly weighed samples of 0.50 g ($\pm$ 0.025 g) was added to a Teflon crucible and digested in a solution of HNO_3: HCl: H_2O_2 (8:1:1 v/v) using microwave (Multiwave 3000, Anton Parr). After digestion, samples were diluted with purified water to 50 mL for the ICP-OES analysis. The detailed information of the aforementioned instruments can be found in our previous publications [4,36–38] or in the supplementary material.

2.4. Biomass CFP experiment

In the CFP experiment, two groups of zeolite catalysts were used in CFP. The first group was the microporous zeolites, including HZSM-5 with two different Si/Al ratios (HZSM-5(25) and HZSM-5(100)) and HY with the Si/Al ratio of 5. The second group was the mesoporous zeolites, including Al-MCM-41 with Si/Al ratio of 25 and USY with Si/Al ratio of 9. These five types of catalysts were labeled as H-5(25), H-5 (100), HY, MCM, and USY, respectively. The basic properties of these catalysts, such as pore structure and acidity, can be found in our previous publication [4].

Py-GC/MS, mainly composed of a CDS-5200 pyroprobe reactor and an Agilent 7890B-5977B gas chromatography/mass spectrometer, was employed as the pyrolysis reactor during the CFP of the raw and wet torrefied bamboo. The feedstock-to-catalyst ratio was varied from 1:1, 1:2, 1:3, 1:4, to 1:5. The feedstock and catalyst were placed in a quartz tube as different layers, and were separated by the quartz wool. Then, the pyrolysis reactor was heated to different pyrolysis temperatures (450, 550, 650, 750, and 850 °C) at a fixed heating rate (20 °C/ms) and a fixed residence time (20 s). The heating program of GC oven was set as starting temperature of 40 °C (kept for 3 min), then heated to 280 °C at 10 °C/min. Other details about the operation conditions of Py-GC/MS can be found in our previous works [7,36,38,39]. The yields of the different compounds in bio-oil were determined by using the semi-quantitative method, and calculated by the ratio of the absolute peak area (a.u.) and sample mass (mg) as shown in Eq. (4).

$$Yield\ of\ compound\ in\ bio-oil\ (a.u.\Big/mg) = \frac{Absolute\ peak\ area\ (a.u.)}{\text{Sample}\ mass\ (\text{mg})} \tag{4}$$

3. Results and discussion

3.1. Effect of WTP process parameters on properties of torrefied products

3.1.1. Mass yield distribution of torrefied products at varying WTP temperatures and durations

Table 1 presents the effects of WTP temperature and duration on the mass distribution of three types of wet torrefied products. Among these three types of torrefied products, the mass yield of the torrefied solid product was much higher than that of torrefied liquid and gaseous products at any WTP condition, indicating that the torrefied solid product was the dominant product during WTP process. The mass yield of wet torrefied solid product reduced from 69.12% to 42.97% with the increase in WTP temperature from 180 °C to 260 °C at a constant WTP duration of 150 min, while the yields of torrefied liquid and gaseous products increased from 27.02% and 3.86–42.43% and 14.6%, respectively. The decrease of the torrefied solid product was attributed to the more severe decomposition of biomass components (e.g., cellulose, hemicellulose, and lignin) at higher WTP temperature via the hydrolysis and deoxygenation reactions [29,40]. Higher percentage of bamboo mass was transformed into the torrefied liquid and gaseous products.

In addition, longer WTP duration also resulted in lower mass yield of

Table 1
Effects of WTP temperature and duration on the mass yields of three types of torrefied products.

WTP temperature (°C)	WTP duration (mins)	Mass yields of torrefied solid, liquid, and gaseous products (wt %)		
		Solid	Liquid	Gaseous
180	30	71.57	25.62	2.81
	60	70.77	26.01	3.22
	90	70.16	26.31	3.53
	120	69.69	26.69	3.62
	150	69.12	27.02	3.86
200	30	67.26	28.78	3.96
	60	67.11	28.91	3.98
	90	66.88	28.94	4.18
	120	64.64	29.87	5.55
	150	63.50	30.46	6.04
220	30	62.35	30.89	6.76
	60	60.74	32.43	6.83
	90	59.89	33.16	6.95
	120	58.73	34.15	7.12
	150	56.10	36.73	7.17
240	30	55.29	37.12	7.59
	60	54.12	38.26	7.62
	90	52.76	39.48	7.76
	120	51.18	40.66	8.16
	150	49.72	41.13	9.15
260	30	47.81	41.15	11.04
	60	46.61	41.65	11.74
	90	45.43	42.16	12.41
	120	43.99	42.21	13.80
	150	42.97	42.43	14.60

torrefied solid product and higher mass yields of torrefied liquid and gaseous products. At the constant WTP temperature of 180 °C, the mass yield of wet torrefied solid product decreased from 71.57% to 69.12% with increase in WTP duration from 30 to 180 min, while the yields of torrefied liquid and gaseous products increased from 25.62% and 2.81–27.02% and 3.86%, respectively. In general, higher WTP temperature and longer WTP time enhanced the thermal decomposition of biomass, and promoted the formation of torrefied gaseous and liquid products. The variation trend of the mass distribution of torrefied products as the function of WTP temperature and duration was also observed by Vople et al. [41] and Tag et al. [42].

WTP temperature and duration could both regulate the mass distribution of three types of torrefied product. In order to evaluate different effects of temperature and duration, the variation rates of the mass yields of three types of torrefied products are presented in Fig. 2, where a positive value means the increase in mass yield, while a negative value indicates the decrease in mass yield. The variation rates of torrefied solid product as a function of WTP temperature and duration were both negative values, indicating a decrease in mass yield with increase in temperature and duration, whereas the variation rates of torrefied liquid and gaseous products presented the opposite variation trend. It is clearly seen that the increase in WTP temperature led to great variation of mass yield compared to the increase in WTP duration. For example, the variation rate of the mass yield of torrefied solid product at 260 °C was in the range of 33.26–37.83% (as shown in Fig. 2 (a1)), whereas the variation rate at 150 min was within a very narrow range of 3.42–10.12% (as shown in Fig. 2 (a2)). Compared to higher WTP duration, higher WTP temperature resulted in lower mass yield. This result suggested that WTP temperature had greater impact on mass distribution of three types of products than WTP duration.

3.1.2. Evolution of basic property of BWT at varying WTP temperatures and durations

Table 2 lists the influence of WTP temperature and duration on the properties of BWT. According to the proximate analysis, the percentage of volatiles in torrefied solid product was greater compared to ash and fixed carbon. In general, the increase in WTP temperature and duration caused a decrease in content of volatiles and ash, but resulted in an increase in content of fixed carbon. For example, compared to the sample BWT-180–30, the content of volatiles and ash compounds decreased from 88.86% and 0.03–57.28% and 0.02% in sample BWT-260–30, respectively, while the content of fixed carbon increased from 11.31% to 42.7%. This result indicated that the carbonization process was promoted at higher WTP temperature and longer WTP duration. It was reported that de-ashing was one of the advantages for WTP. After WTP, the de-ash rate reached the maximum value of 99.12%. The inorganic components (e.g. potassium, calcium, sodium, magnesium) in ash were dissolved in the water solution, and removed from the torrefied solid product [22,31].

The carbon content in torrefied bamboo increased with the increase in WTP temperature and duration, while the oxygen content decreased, leading to a sharp decline of the O/C ratio from 0.77 to 0.25. Furthermore, the deoxygenation rate increased from 6.91% for sample BWT-180–30–49.36% for sample BWT-260–150. Thus, WTP could remove half of the oxygen element in biomass by a series of deoxygenation reactions during the thermal decomposition process of hemicellulose and cellulose. Hoekman et al. [27] and Stemann et al. [26] reported that the oxygen element in raw feedstock was transferred into the torrefied non-condensable gaseous products (e.g., CO_2 and CO) and the condensable liquid product (e.g., acids and anhydro-sugars). The removal of oxygen and the enhancement of carbon in torrefied solid product at severe WTP condition caused a remarkable increase in the HHV, from 18.85 MJ/kg for raw feedstock to 25.31 MJ/kg for BWT-260–150.

3.1.3. Demineralization rates of AAEMs at varying WTP temperatures

As discussed in the foregoing paragraphs, WTP temperature was the more dominant influencing factor compared to WTP duration. Thus, the influence of WTP temperature on the properties of BWT was investigated at a constant WTP duration of 150 min in the following sections. K, Ca, Na, and Mg are the four typical AAEMs in lignocellulosic biomass.

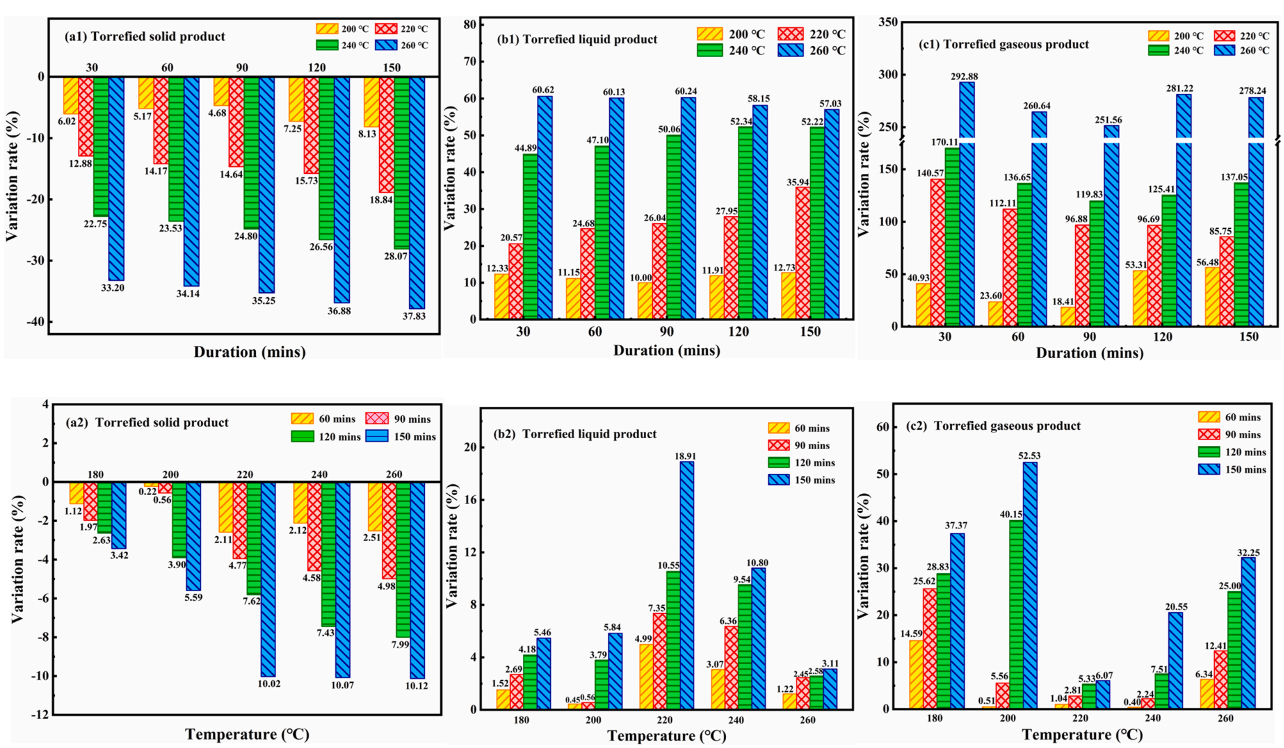

Fig. 2. Variation rate of the mass yields of torrefied solid, liquid, and gaseous products as the function of WTP temperature (a1, b1, c1) and duration (a2, b2, c2) (compared to sample of BTW-180–30, a positive value or negative value indicated the increase or decrease in mass yields, respectively).

Table 2
Effects of WTP temperature and duration on the properties of raw bamboo and BWT.

Torrefied solid product	Proximate analysis (wt%)			Ultimate analysis (wt%)						HHV (MJ/kg)	Energy yield （%）	Deoxygenation rate (%)	De-ash rate (%)
	Ash	Vol.[1]	F.C.[2]	C	H	O	N	S	O/C				
Raw	1.14	82.96	15.9	46.12	6.11	47.47	0.05	0.25	0.77	18.85	100.00	/	/
BWT-180–30	0.03	88.66	11.31	49.03	6.56	44.19	0.02	0.20	0.68	18.42	69.94	6.91	97.37
BWT-180–60	0.02	87.65	12.33	49.36	6.63	43.83	0.03	0.15	0.67	18.58	69.76	7.67	98.25
BWT-180–90	0.01	87.52	12.47	49.70	6.36	43.79	0.03	0.12	0.66	18.70	69.60	7.75	99.12
BWT-180–120	0.01	87.24	12.75	50.57	6.48	42.84	0.02	0.09	0.64	18.66	68.99	9.75	99.12
BWT-180–150	0.01	86.55	13.44	50.95	6.45	42.50	0.03	0.07	0.63	18.80	68.94	10.47	99.12
BWT-200–30	0.01	85.26	14.73	51.98	6.37	41.60	0.03	0.02	0.60	19.30	68.87	12.37	99.12
BWT-200–60	0.01	84.76	15.23	52.19	6.31	41.40	0.02	0.08	0.59	19.33	68.82	12.79	99.12
BWT-200–90	0.01	83.08	16.91	52.54	6.34	40.95	0.02	0.15	0.58	19.38	68.76	13.73	99.12
BWT-200–120	0.01	82.04	17.95	52.90	6.24	40.78	0.02	0.06	0.58	19.54	67.01	14.09	99.12
BWT-200–150	0.01	81.9	18.09	53.09	6.25	40.52	0.03	0.11	0.57	19.57	65.93	14.64	99.12
BWT-220–30	0.02	81.4	18.58	53.54	6.28	40.15	0.03	0.00	0.56	19.63	64.93	15.42	98.25
BWT-220–60	0.02	77.4	22.58	54.07	6.18	39.62	0.03	0.10	0.55	19.70	63.48	16.54	98.25
BWT-220–90	0.01	77.24	22.75	54.53	5.96	39.48	0.03	0.00	0.54	19.96	63.42	16.83	99.12
BWT-220–120	0.01	75.14	24.85	55.09	5.79	39.00	0.03	0.09	0.53	20.33	63.34	17.84	99.12
BWT-220–150	0.01	71.46	28.53	55.54	6.28	38.15	0.03	0.00	0.52	21.28	63.33	19.63	99.12
BWT-240–30	0.03	70.45	29.52	57.71	5.88	36.28	0.02	0.11	0.47	21.46	62.95	23.57	97.37
BWT-240–60	0.02	66.42	33.56	62.57	5.83	31.51	0.03	0.06	0.38	21.89	62.85	33.62	98.25
BWT-240–90	0.02	61.83	38.15	64.00	5.58	30.38	0.03	0.01	0.36	22.34	62.53	36.00	98.25
BWT-240–120	0.01	56.36	43.63	67.77	5.47	26.74	0.02	0.00	0.30	23.01	62.47	43.67	99.12
BWT-240–150	0.01	55.49	44.5	68.23	5.42	26.33	0.01	0.01	0.29	23.44	61.83	44.53	99.12
BWT-260–30	0.02	55.28	44.7	68.84	5.46	25.61	0.03	0.06	0.28	24.54	62.24	46.05	98.25
BWT-260–60	0.02	54.46	45.52	70.04	5.42	24.51	0.02	0.01	0.26	25.10	62.06	48.37	98.25
BWT-260–90	0.01	54.15	45.84	70.21	5.38	24.30	0.02	0.09	0.26	25.29	60.95	48.81	99.12
BWT-260–120	0.01	53.72	46.27	70.53	5.35	24.11	0.01	0.00	0.25	25.23	58.88	49.21	99.12
BWT-260–150	0.01	53.31	46.68	70.56	5.39	24.04	0.01	0.00	0.25	25.31	57.70	49.36	99.12

[a] Vol. means volatiles;
[b] F.C. means fixed carbon

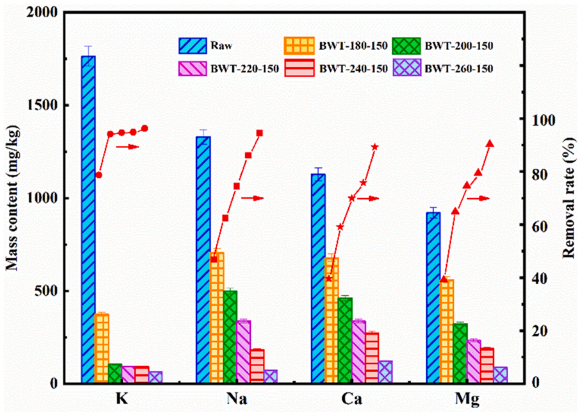

Fig. 3. Contents of AAEMs in torrefied solid product as a function of WTP temperature.

As presented in Fig. 3, the contents of these four types of AAEMs in the raw bamboo were in the order of K (1763.34 mg/kg) > Na (1330.12 mg/kg) > Ca (1128.44 mg/kg) > Mg (921.42 mg/kg). The increase in WTP temperature from 180 to 260 °C caused a sharp decrease in the contents of all four types of AAEMs. Specifically, the content of K decreased from 375.14 to 65.35 mg/kg, Na from 707.73 to 72.63 mg/kg, Ca from 680.15 to 121.59 mg/kg, and Mg from 560.09 to 89.13 mg/kg.

The difference in the removal rates among these four types of AAEMs was strongly related to their chemical bound forms with the organic substance in biomass. The AAEMs are mainly composed of the following three groups: (i) water-soluble metals (e.g., KCl, NaCl, K_2CO_3, Na_2CO_3, $MgCO_3$), (ii) ion-exchanged metals (the metals organically associated with biomass matrix), and (iii) the acid-soluble metals (mainly the alkaline earth metals) [43]. At the low WTP temperature of 180 °C, the water-soluble metals were removed from biomass. The demineralization rates at 180 °C were in the order of K (78.73%) > Na (46.79%) > Ca (39.73%) > Mg (39.21%), suggesting that K was the easiest metal to be removed from biomass even at low WTP temperature, because K mainly existed in the water-soluble form. In the mild WTP temperature range of 200–240 °C, the removal rates of these four AAEMs increased slowly. At this stage, the acidity of reaction solution increased because of the formation of acids derived from pyrolysis of cellulose and hemicellulose [27,29]. More acid-soluble and ion-exchanged AAEMs released from the torrefied solid product. At high WTP temperature (260 °C), the demineralization rates continued to increase, and reached their maximum values of 96.29% (K), 94.54% (Na), 90.33% (Mg), and 89.22% (Ca). Zhang et al. obtained the similar conclusion that higher WTP temperature resulted in higher removal rate of AAEMs from rice husk, where the maximum demineralization rates of were 92.67% (K), 92.19% (Na), 84.71% (Mg), and 78.92% (Ca) [31]. In conclusion, WTP was found to be an effective approach to remove AAEMs from lignocellulosic biomass. In addition, K and Na were removed more easily from biomass compared to Ca and Mg.

3.1.4. XRD and FTIR analyses of wet torrefied solid product

Fig. 4(a) presents the evolution of crystallinity of torrefied solid product as a function of WTP temperature. The crystallinity index (CrI) is sensitive to the content of cellulose in lignocellulosic biomass [37,44]. Two typical crystalline structures of cellulose, namely, cellulose I_α (triclinic) and cellulose I_β (monoclinic), were clearly observed at 2θ of 16° and 22 °. First, the CrI increased from 36.75% for raw feedstock to 52.17% for sample BWT-220–150. In the lower WTP temperature, the majority of hemicellulose, a small part of lignin, and the amorphous region of cellulose were degraded, while the crystalline region of cellulose was not damaged, leading to the increase in CrI [45]. However, the CrI sharply decreased from 52.17% for sample BWT-220–150–7.14% for sample BWT-260–150 because of the severe destruction of crystalline region of cellulose. Majority of cellulose was degraded at the severe WTP condition, and converted to the torrefied non-condensable and condensable products. The evolution trend of CrI at varying WTP temperatures was in agreement with the result reported by Xue et al. [45] using corncob as feedstock.

Fig. 4(b) illustrates the evolution of chemical functional groups of torrefied solid product as a function of WTP temperature. The characteristic absorption bands of torrefied solid product were identified based on the previous works [37,46,47]. The band at 3461 cm^{-1} was caused by the stretching vibration of O–H from the hydroxyl/carboxyl groups, and the band at 2841 cm^{-1} was attributed to the stretching vibration of C–H from aliphatic groups. The decrease in intensity of these two bands at higher WTP temperature suggested that these chemical functional groups was cracked because of the thermal decomposition of hemicellulose and cellulose at severe WTP condition. The breaking of O–H and C–H bonds led to the formation of H_2O and CH_4, respectively [27]. Furthermore, higher WTP temperature also resulted in a decrease in intensity of the stretching vibration peak of C=O at 1777 cm^{-1} because of the decarboxylation and decarbonylation reactions, during which this functional group was transferred into CO_2 and CO in the torrefied gaseous product [48]. The C=C peak at 1690–1450 cm^{-1} showed only slight variation. In the fingerprint region of 1475–1000 cm^{-1}, the stretching vibration bands of C–O and C–H in aliphatic $-CH_3$ or phenolic–OH decreased, suggesting the breakage of methyl and hydroxyl groups [49].

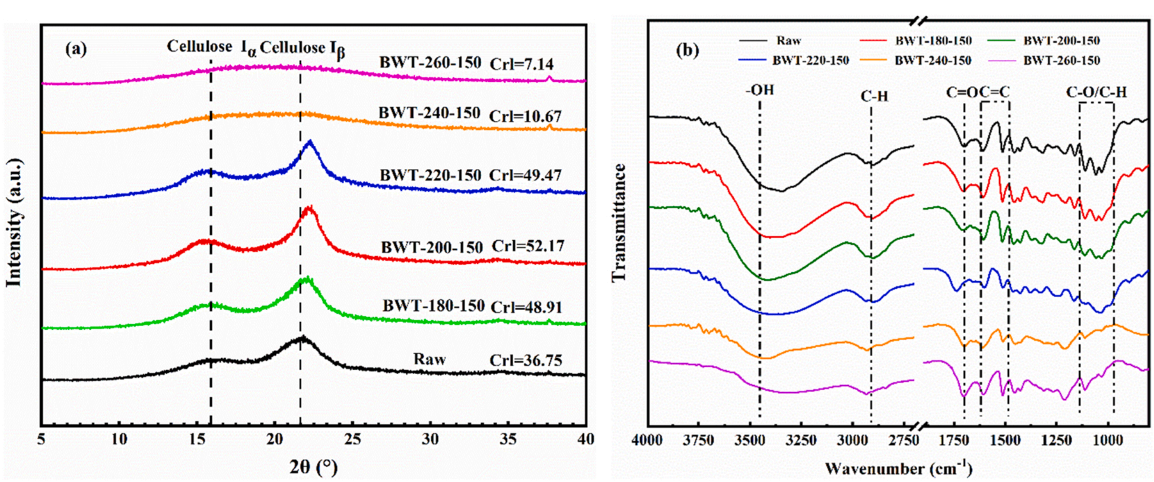

Fig. 4. XRD and FTIR analyses of raw and wet torrefied biomass samples.

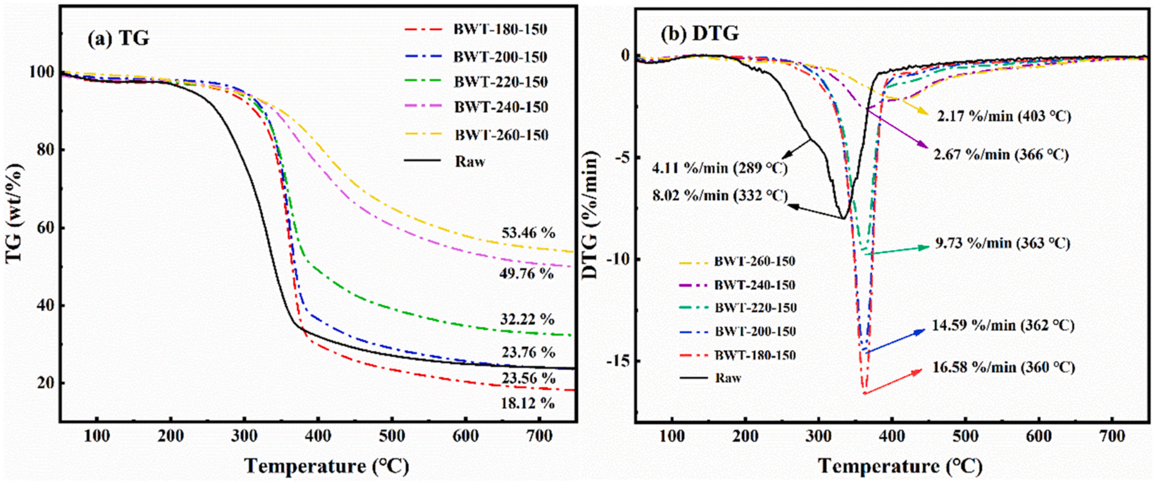

Fig. 5. TG and DTG curves of the fast pyrolysis process of raw bamboo and BWT samples.

3.1.5. TGA analysis of raw and wet torrefied solid product

The thermal decomposition behaviors of the raw bamboo and BWT are shown in Fig. 5. The TG curves exhibited an increasing trend of the residue mass from 18.12% to 53.46% with the increase in WTP temperature from 180 to 260 °C, because the volatile content in torrefied solid product decreased at higher WTP temperature (as listed in Table 2) [50]. It is worth noting that the residue mass of BWT-180–150 was lower than the raw sample. After wet torrefaction at 180 °C, the chemical structure recalcitrance of bamboo reduced. Therefore, the substance in torrefied bamboo was easier to be degraded during TGA, resulting in a decrease of the residue mass after pyrolysis. For the raw feedstock, two mass loss peaks at 289 and 332 °C, namely the shoulder peak and the sharp peak, were observed from the DTG curve. The shoulder peak at lower pyrolysis temperature was attributed to the thermal decomposition of the most reactive hemicellulose, while the sharp peak at higher pyrolysis temperature was derived from the thermal decomposition of cellulose.

However, the shoulder peak disappeared after wet torrefaction pretreatment, because majority of hemicellulose was degraded in the WTP process. Aguado et al. reported that hemicellulose was almost entirely degraded when the WTP temperature was 250 °C [49]. The mass loss rate at the sharp peak gradually decreased from 16.58%/min to 2.09%/min with the increase in WTP temperature from 180 to 260 °C, which was attributed to the decrease in cellulose content in torrefied solid product. Aguado et al. also found that the cellulose content reduced from 49% to 18% with the increase in WTP temperature from 175 to 250 °C [49]. In addition, the sharp peak moved towards higher pyrolysis temperature because of the increase in thermostability of wet torrefied solid product at higher WTP temperature, which was also demonstrated by Kamonwat et al. [29] and Li et al. [51].

3.2. CFP of raw and wet torrefied solid product

3.2.1. Effect of WTP temperature on CFP

Fig. 6 presents the influence of WTP temperature on compound distribution in bio-oil. The bio-oil compounds can be divided into three groups, namely the aromatics, the oxygenates, and the aliphatics. Furthermore, the aromatics were composed of the monocyclic aromatic hydrocarbons (MAHs, e.g., toluene, xylene, and benzene) and the polycyclic aromatic hydrocarbons (PAHs, e.g., naphthalene, biphenyl, and phenanthrene). It was clearly observed that the yield of the oxygenates remarkably reduced from 19.66×10^7 to 4.99×10^7 a.u./mg with the increase in WTP temperature from 180 to 260 °C. As stated in Section 3.1.2, WTP was an effective deoxygenation approach which removed 49.36% of oxygen element at the highest WTP temperature. The decrease in content of oxygen element in torrefied bamboo resulted in a decrease in content of oxygenates in bio-oil as a matter of course. Zhang et al. studied the effect of WTP on the compounds of bio-oil from

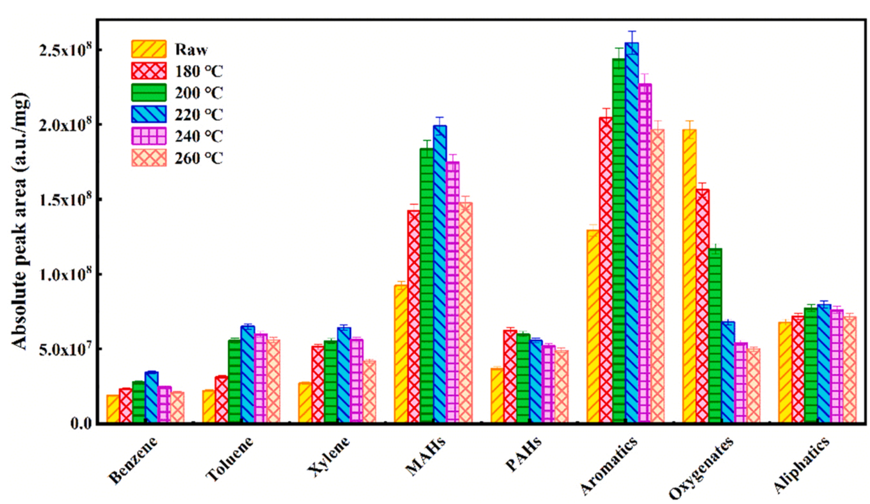

Fig. 6. Effect of WTP temperature on the compound distribution in bio-oil (CFP temperature = 850 °C, catalyst = HZSM-5(25), catalyst-to-biomass mass ratio = 3:1).

fast pyrolysis of rice husk, and found that the relative content of oxygenates (including acids, ketones, furans, and phenols) in bio-oil decreased from 50.3% to 36.2% with the increase in WTP temperature from 150 to 240 °C [31]. Zheng et al. also found that the mass yield of the oxygenates (including ketones, esters, acids, and furans) in bio-oil reduced from 6.8 $_{w.t.}$% to 4.6 $_{w.t.}$% with the increase in WTP temperature from 160 to 190 °C [50].

Furthermore, the yield of aromatics increased at first from 12.94×10^7 to 25.46×10^8 a.u./mg with the increase in WTP temperature from 180 to 220 °C, then decreased gradually to 19.67×10^7 a.u./mg at 260 °C. The results demonstrated that the moderate WTP temperature (220 °C) provided the highest yield of aromatics rather than the highest WTP temperature (260 °C). First, at the mild WTP temperature (180–220 °C), majority of hemicellulose was degraded, leading to improved quality of biomass, such as lower O/C ratio and higher deoxygenation rate. Moreover, the intrinsically recalcitrant organic matter derived from the linkages among the three major components (lignin, hemicellulose, and cellulose) in bamboo was reduced after WTP under mild condition [52]. These two reasons likely caused the increase in content of aromatics after mild WTP. At the severe WTP condition (220–260 °C), cellulose started to decompose, and about half of cellulose was degraded at the WTP temperature of 260 °C. Lu et al. [53] studied the fast pyrolysis mechanism of cellulose, and concluded that cellulose was converted into anhydrosugar and its derivatives (e.g. levoglucosan (LGA), levoglucosenone (LGO), or 1,4:3,6-dianhydro-α-D-glucopyranose) by a series reactions, such as the breakage of glycosidic bond, rearrangement reaction, and dehydration reaction. Then, these intermediate compounds continued to transform into furans by further deoxygenation reactions. According to the hydrocarbon pool mechanism, furans were the key synergistic reactant in Diels-Alder reaction to produce aromatics during biomass CFP process [39]. The decrease in content of cellulose content at severe WTP condition resulted in lower concentration of furans intermediates in pyrolysis vapor, thus leading to the decrease of aromatics. Furthermore, the lignin content increased remarkably at higher WTP temperature which was more favorable for the formation of bio-char through the polycondensation reaction rather than bio-oil.

The yield of MAHs (e.g. toluene, benzene, and xylene) first increased with the increase in WTP temperature from 180 to 220 °C, and then decreased at higher WTP temperature. This evolution pattern was the same as that of the total aromatics. The MAHs could be produced by the following pathways on the acid sites of zeolite catalyst during CFP process, e.g., aldol condensation between two carbonyl compounds, ketonization reaction between two acids, oligomerization and Diels-Alder reactions between oxygenates and light olefins [5]. Therefore, oxygenates were the intermediate reactants in the majority of the reaction pathways. Reasonable content of oxygenates in pyrolysis vapor is necessary for the formation of MAHs. Therefore, the WTP temperature of 220 °C was considered the optimal temperature to produce MAHs.

3.2.2. *Effect of zeolite catalyst on CFP*

Fig. 7 presents the influence of different types of zeolite catalyst on compound distribution in bio-oil. Among the five types of zeolite catalysts, the microporous zeolites (H-5(25), H-5(100), and HY) produced higher yield of aromatics and lower yield of oxygenates compared to the mesoporous zeolites (MCM and USY). The pore size of micropore zeolite was about 5.2–5.7 Å, which was similar to the kinetic diameter of MAHs (benzene, toluene, and xylene) [4]. Therefore, the unique pore size had the ability of shape selectivity to trap more oxygenates into the channel of zeolite and convert them into MAHs [39,54]. Ma et al. [4] and Shen et al. [34] compared the yield of light aromatics from lignin CFP between the microporous zeolite and mesoporous zeolite. This results also confirmed that the microporous zeolite presented higher yield of aromatics due to its shape selectivity towards reactants and products during biomass CFP.

Furthermore, H-5(25) presented the highest yield of aromatics among the three types of mesoporous zeolite (H-5(25), H-5(100), and HY). According to our previous publication [4], the total acidity of H-5 (25) was 1.147 mmol/g, which was much higher than that of H-5 (100) of (0.315 mmol/g) and HY (1.015 mmol/g). This result suggested that acidity of zeolite was an important factor influencing on the yield of aromatics under the circumstance of similar pore size. Kim et al. studied the influence of Si/Al ratio of HZSM-5 on the yield of aromatics [55], and found that lower Si/Al ratio resulted in higher yield of aromatics due to the higher acidity of HZSM-5. The higher acidity of zeolite led to stronger deoxygenation and aromatization ability during biomass CFP process. Therefore, in this work, the catalytic performance of H-5(25) was better than that of H-5(100). In addition, the increase in content of aromatics was at the expense of the oxygenates due to the deoxygenation reaction. Similar to the total aromatics, the maximum yields of single BTX (benzene, toluene, and xylene) were also obtained by employing H-5(25) as catalyst. Among these three types of MAHs, the content of toluene was slightly higher than that of benzene and xylene, because toluene with one methyl group on benzene ring was easier to be formed during biomass CFP process [4,54].

3.2.3. *Effect of catalyst-to-biomass ratio on CFP*

Fig. 8 presents the influence of the catalyst-to-biomass ratio (C/B) on

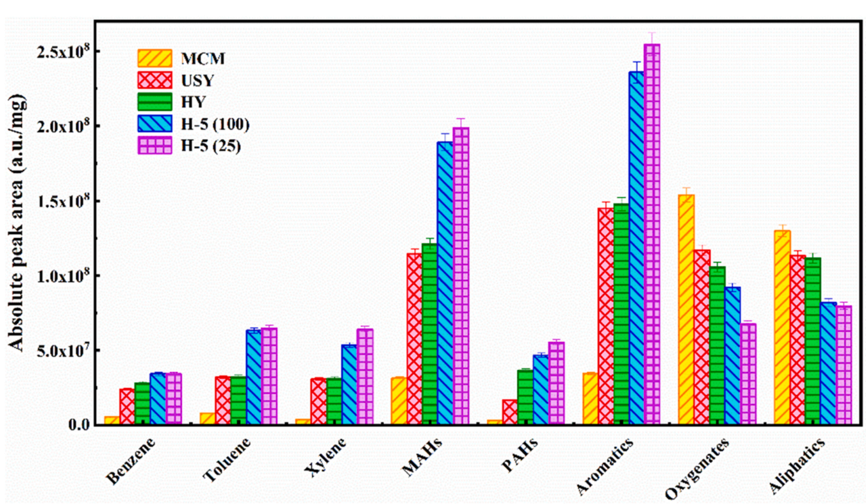

Fig. 7. Effect of different types of zeolite catalyst on the compound distribution in bio-oil (WTP temperature = 220 °C, CFP temperature = 850 °C, catalyst-to-biomass mass ratio = 3:1).

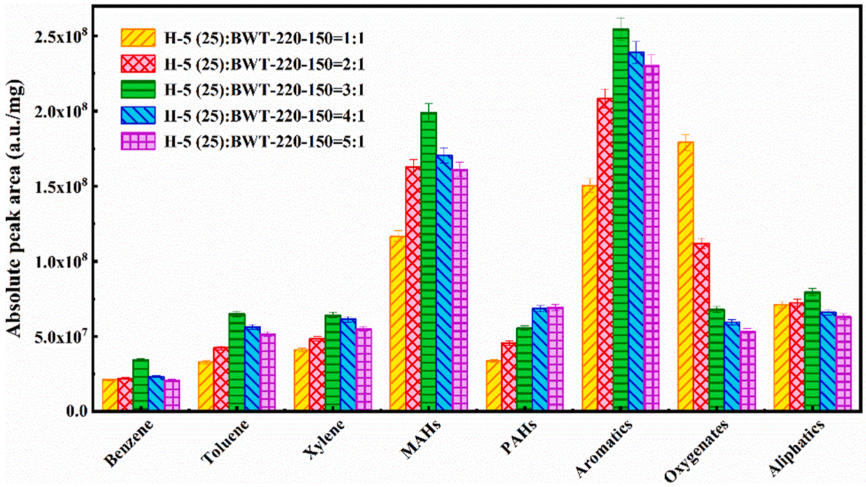

Fig. 8. Effect of catalyst-to-biomass ratio on the compound distribution in bio-oil (WTP temperature = 220 °C, CFP temperature = 850 °C, catalyst = HZSM-5(25)).

compound distribution in bio-oil. The yield of aromatics increased first from 15.06 × 10^7 to 25.46 × 10^7 a.u./mg with the increase in C/B ratio from 1:1–1:3, and then decreased slightly to 23.06 × 10^7 a.u./mg at the highest C/B of 1:5. At low C/B ratio (1:1–1:3), the increase in C/B ratio could provide more acid sites for the conversion of the oxygenates to aromatics through the deoxygenation and aromatization reactions during CFP process [4]. The increase in yield of aromatics was accompanied by the consumption of oxygenates, resulting in a continuous decrease in content of oxygenates. However, at high C/B ratio (1:4–1:5), excessive catalyst used in biomass CFP promoted the conversion of aromatics into other gaseous products with small molecular weight or the coke formation [56]. This result was also concluded by Zhang et al. [57] and Paasikallio et al. [58]. Zhang et al. [57] studied the evolution of aromatics at varying C/B ratios from 1:1–1:10, and found that the yield of aromatics reached its maximum value at the C/B of 3:1. Paasikallio et al. [58]. also reported that higher C/B ratio enhanced the cracking effect of the catalyst, converting more aromatics into other gaseous products and water.

The aromatics were composed of MAHs and PAHs. The evolution of the yield of MAHs at varying C/B ratios was similar to that of the total aromatics, where the yield of MAHs reached its maximum value at the moderate C/B ratio of 3:1. In contrast, the yield of PAHs increased gradually from 3.38 × 10^7 to 6.94 × 10^7 a.u./mg with the increase in C/B ratio from 1:1–1:5. As shown in the supplementary material, naphthalene, naphthalene derivatives (e.g., 1-methyl-naphthalene, 2-ethyl-naphthalene, and 1, 3-dimethyl-naphthalene), anthracene, and phenanthrene were the dominant compounds in PAHs. Excessive catalyst used in biomass CFP increased the retention time of the reactants in the channel of zeolite. Therefore, the MAHs became the precursor for the formation of PAHs and coke by polycondensation reaction [58]. Luo et al. [59] and Yang et al. [56] found that excessive catalyst promoted the secondary polycondensation reaction for converting light aromatics into PAHs or coke. In conclusion, the use of a reasonable amount of catalyst in biomass CFP can avoid the formation of PAHs at the expense of light aromatics.

3.2.4. *Effect of pyrolysis temperature on CFP*

Fig. 9 presents the influence of the CFP temperature on compound distribution in bio-oil. The yield of aromatics gradually increased from 20.3 × 10^7 to 25.46 × 10^7 a.u./mg with the increase in CFP temperature from 450 °C to 850 °C. However, the yield of oxygenates gradually reduced from 18.46 × 10^7 to 6.79 × 10^7 a.u./mg. Therefore, the evolution of the yield of aromatics showed the opposite trend to the yield of oxygenates. This result suggested that the increase in yield of aromatics was most probably at the expense of oxygenates. Higher CFP temperature favored of the breakage of inter-coupling linkages among the three-

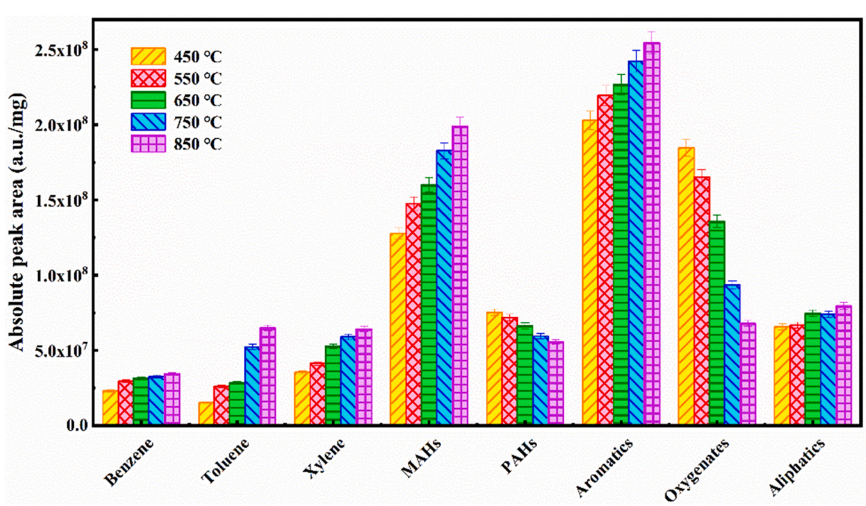

Fig. 9. Effect of CFP temperature on the compound distribution in bio-oil (WTP temperature = 220 °C, catalyst-to-biomass mass ratio = 3:1, catalyst = HZSM-5 (25)).

main components of biomass (lignin, hemicellulose, and cellulose) or the thermal decomposition of three-main components themselves, which released more oxygenating intermediate during pyrolysis process [60]. These oxygenates can be trapped in the pore structure of HZSM-5 (25) catalyst, and converted into aromatics on the acid sites of zeolite catalyst [4,54]. The increase in the yield of aromatic with increase in temperature was consistent with the conclusions of Huang et al. [4] and Kim et al. [55]. Huang et al. [4] reported that higher temperature promoted the breakage of the inter-linkage between the basic structure units of lignin (G, S, and P type phenols) and the oxygenated functional

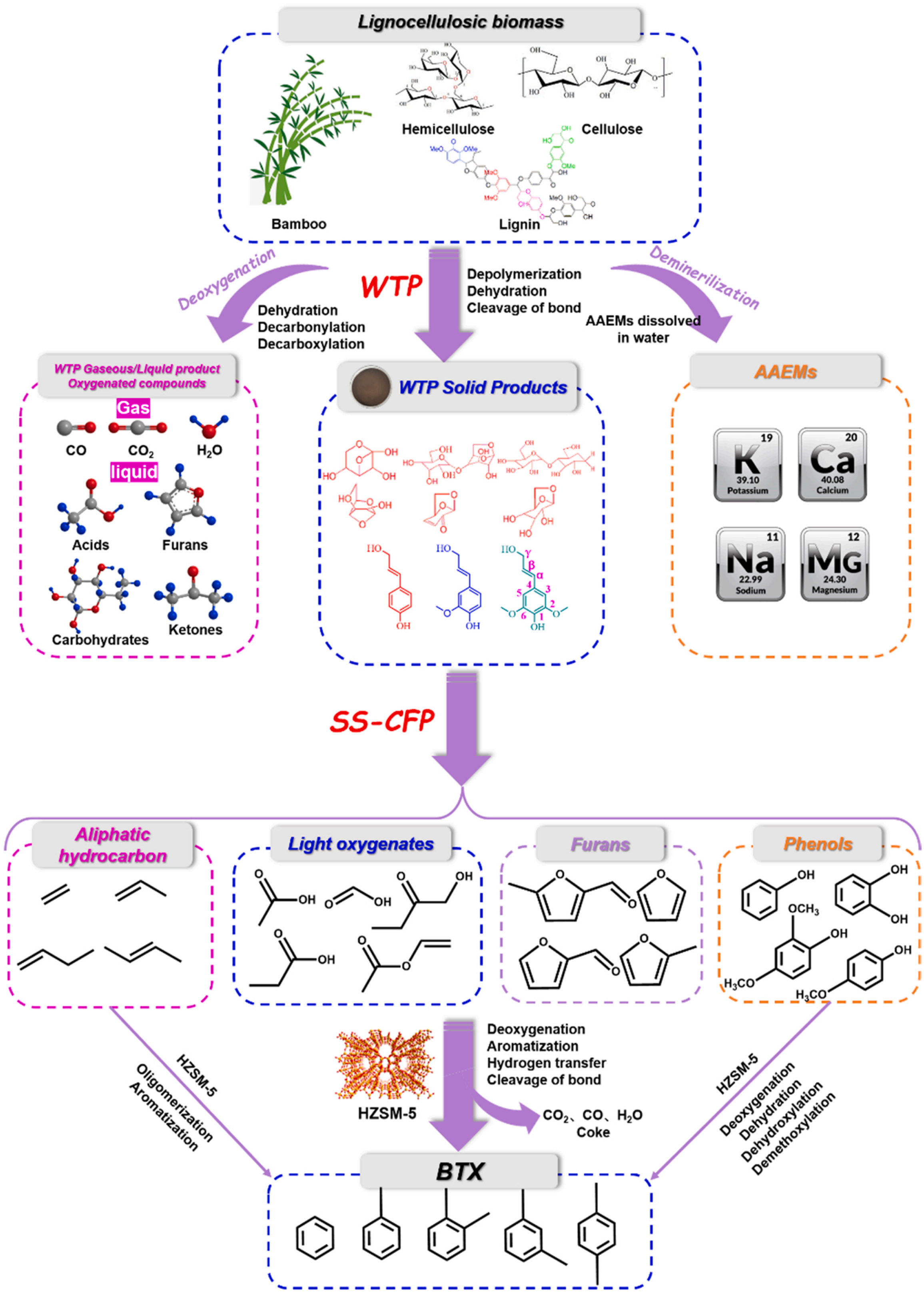

Fig. 10. Formation mechanism of bio-aromatics by the combined approach of WTP and CFP.

groups (—OCH_3 and —OH) on the benzene structure, thus resulting in an increase in yield of aromatics. Furthermore, higher pyrolysis temperature reduced the activation energy of oxygenated intermediates diffused into the pore structure of HZSM-5(25) catalyst for conversion into aromatics [54].

It is worth noting that the yield of MAHs gradually increased from 12.77×10^7 to 19.9×10^7 a.u./mg with the increase in CFP temperature, while the yield of PAHs decreased from 7.53×10^7 to 5.57×10^7 a.u./mg. This result indicated that higher pyrolysis temperature was more favorable for the formation of MAHs compared to PAHs, because higher CFP temperature promoted the second cracking of PAHs. Among the three types of MAHs, the yields of toluene and xylene were slightly higher than that of benzene at high CFP temperature. At the CFP temperature of 800 ºC, the yields of these three compounds were in the order of toluene > xylene > benzene. Huang et al. studied the effect of CFP temperature on the yield of MAHs, and also found that toluene formation was more favored at higher CFP temperature [4]. Gu et al. [60] studied the influence of WTP pretreatment on the CFP of tobacco stems, and also found that toluene was the dominant monocyclic component in aromatics. In conclusion, higher CFP temperature promoted the formation of MAHs, especially the monocyclic toluene.

4. Formation mechanism of bio-aromatics by the combined approach of WTP and CFP

In this work, the formation mechanism of bio-aromatics by the combined approach of WTP and CFP is presented in Fig. 10. In the WTP process, part of the oxygen element and majority of the AAEMs are removed from the bamboo. The three-major components (hemicellulose, cellulose, and lignin) in bamboo can be degraded during WTP process. The oxygen element in bamboo is removed in the form of the torrefied gaseous and liquid products. Non-condensable CO_2 and CO are the dominant oxygen carriers in torrefied gaseous product [48]. Organic acids (e.g., formic acid, acetic acid, and lactic acid) and sugars (e.g., xylose, glucose, fructose, and levoglucosan) are the oxygen carriers in torrefied liquid product [27,29]. The AAEMs were removed in different forms of metal compound. At the low WTP temperature, the K and Na are the easiest metals to be removed from biomass in the water-soluble form, such as KCl, K_2CO_3, NaCl, and Na_2CO_3. At the severe WTP temperature, more acid-soluble and ion-exchanged AAEMs (Mg and Ca) are released from bamboo, which is attributed to the increase in the acidity of the reaction solution because of the increasing concentration of different acids produced at higher WTP temperature.

In the CFP process, the three-major components (cellulose, hemicellulose, and lignin) in torrefied bamboo continue to degrade, and converted to pyrolyzed gaseous intermediates. The degradation of cellulose starts from the breakage of the glycosidic bond and the opening reaction of pyran ring, and generates different types of anhydro-sugars (e.g. levoglucosan) and its derivatives (e.g. 1,4:3,6-dianhydro-α-D-glucopyranose and 1, 6-anhydro-β-D-glucofuranose). Then, these anhydrosugars are converted into furans and other light oxygenates (e.g., acids, ketones, and aldehydes) through dehydration, retro-aldol condensation, and fragmentation reactions. Thermal degradation of hemicellulose can be started from the breakage of glycosidic bond where the double-hydrated xylose, xylose, and 1,4-dehydration-D-xylopyranose (ADX), furfural (FF) are produced. Then, the ring-opening reaction of xylose and ADX occurs, and is converted into hydroxyacetaldehyde (HAA), hydroxyacetone (HA), and other light oxygenates. In addition, the breakage of side-chain of xylan leads to the formation of several oxygenates (e.g. acetic acid, formaldehyde, and hydroxyaceton). Pyrolysis of lignin is rather easily started from the cracking of the aryl ether bond and C—C bond linked between the phenylpropane units, and is converted into several phenolic compounds (e.g., phenols, guaiacols, syringols, and catechols). Several olefins are produced by the dihydroxylation and decarbonylation reaction on the side-chain of phenylpropane units.

In this work, several species of zeolite catalysts have been screened during CFP of torrefied bamboo, i.e., H-5(25), H-5(100), HY, MCM and USY. H-5(25) presented highest yield of aromatics which was attributed to its well-balanced acidity and unique pore size (5.2–5.7 Å). In the present of H-5(25), the aforementioned light oxygenates derived from the primary pyrolysis of the three-major components (cellulose, hemicellulose, and lignin) diffuse into the micropore of HZSM-5, and converted into the light aromatics in the hydrocarbon pools. The olefins are converted to the light aromatics through the oligomerization and aromatization reactions. Furthermore, the light oxygenates (e.g. furans, acids, aldehydes, ketones, and phenols) are converted to the light aromatics via the deoxygenation reactions (dihydroxylation, decarbonylation, decarboxylation, dimethoxy), oligomerization reaction, and aromatization reaction.

5. Conclusion

A combined approach of WTP and CFP was employed to improve the yield of bio-aromatics from bamboo. WTP temperature had a more significant influence on the mass yields of torrefied products compared to WTP duration. The maximum deoxygenation rate was 49.36% at WTP conditions of 260 °C and 150 min, and the maximum demineralization rate was in the order of 96.29% (K) > 94.54% (Na) > 90.33% (Mg) > 89.22% (Ca). The maximum yield of aromatics (25.46×10^7 a.u./mg) was obtained at the WTP temperature of 220 °C, biomass-to-catalyst ratio of 3:1, and CFP temperature of 850 °C. Toluene was the more favored MAHs formed during CFP compared to xylene and benzene.

CRediT authorship contribution statement

Zhouyang Hu: Investigation, Writing – original draft. **Liang Zhu:** Investigation, Writing – original draft. **Hongyi Cai:** Visualization, Software. **Ming Huang:** Visualization, Software. **Jie Li:** Visualization, Software. **Bo Cai:** Visualization, Software. **Dengyu Chen:** Conceptualization, Supervision. **Lingjun Zhu:** Conceptualization, Supervision. **Youyou Yang:** Funding acquisition, Writing – review & editing. **Zhongqing Ma:** Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jaap.2022.105818.

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